

Finite-Sample Misranking in Stochastic Decision Systems: A Paired-Sampling Framework for Quantifying Decision Reliability

Abstract

[Final abstract to be inserted here.]

Research Question. How many samples are required to reliably identify the superior rule among two stochastic decision systems, and how does paired sampling reduce the probability of finite-sample misranking?

Keywords: finite-sample misranking; paired sampling; stochastic decision systems; Monte Carlo simulation; variance reduction; sample-size estimation

1. Introduction

In stochastic decision systems, a strategy can be superior in expectation yet look worse in a finite-sample experiment. This happens because random variation can temporarily overwhelm the true mean advantage, resulting in what this study refers to as a “misranking”—a phenomenon where the inferior strategy outperforms the superior strategy in the short run. This is because expected superiority reflects underlying probability distributions, while observed superiority usually depends on sample size. As sample size increases, the law of large numbers causes observed averages to converge toward their expected values, making such reversals progressively less likely. Hence, this study uses the term “misranking” specifically in the finite-sample context. The central question, therefore, is not which rule is superior in expectation but how much evidence is required before a ranking can be trusted.

When the true difference between two rules is small compared to the variability of their results, the probability of misranking can still be substantial—even when the superior rule is known. Increasing sample size helps reduce this uncertainty. However, how quickly results become more reliable depends mainly on two quantities: the magnitude of the true mean advantage and the variance of the difference between the competing rules.

For better comparison, a matched-pair study design can be more efficient, as here, all strategies are exposed to the same stochastic environment and its conditions; as these shared conditions induce positive covariance between their outcomes, environmental variation is partly cancelled when the difference between the rules is computed, reducing the variance of the comparison. Therefore, this study hypothesises that matched evaluation can reduce both the probability of finite-sample misranking and the number of observations required to reach a desired level of reliability.

Blackjack serves as the primary high-variance case study for the hypothesis to be tested, although the underlying problem is significantly broader. Similar ranking errors can arise whenever competing policies are evaluated under stochastic demand, service conditions, or other random environments. The purpose of this work is therefore to develop a general Paired Finite-Sample Misranking Estimator (PFME) rather than a blackjack-specific simulation.

Hence, the four primary contributions of this study are:

1. formalising finite-sample misranking using paired differences;
2. deriving relationships for reversal probability and the sample size required for a target reliability level;
3. quantifying the variance reduction that can be achieved through paired evaluation;
4. validating the framework across multiple stochastic domains using empirical, analytical, and exact benchmarks.

2. Literature Review

The problem of identifying a superior strategy from multiple noisy observations appears in many stochastic systems. Blackjack serves as an useful high variance case study, but the mathematical problem of optimal blackjack play itself is solved and mature. Baldwin, Cantey, Maisel, and McDermott [1] derived an optimal playing strategy as early as 1956 by maximizing mathematical expectation under specified game rules, and later work extended blackjack analysis to betting, card counting, deck composition, and rule-specific expected returns [2]; the strategy of optimal play was later named the “Basic Strategy”. Van der Genugten [3], for example, combined analytical and simulation methods to study basic, optimal, and winning strategies under casino rules. More recent work has continued to use Monte Carlo simulation to compare blackjack strategies and betting systems. Vaidyanathan’s 2014 undergraduate thesis similarly used Monte Carlo methods to compare blackjack strategies, reaching the conclusion that Basic Strategy minimized house edge and gave highest baseline returns.

Nonetheless, the broader statistical issue goes beyond blackjack. In the field of operations research, ranking and selection (R&S) deals with how simulations can help to find the best from noisy stochastic systems and maintain the probability of wrong selection at a low level. According to Kim and Nelson [6], R&S refers to a variety of experimental design methods the goal of which is to find the best system from competing stochastic systems, by specifying an acceptable level of probability of correct selection and to determine the amount of simulation necessary to achieve this goal. Moreover, the current state of the research has resulted in the creation of general, sequential, Bayesian, and finite sample methods including approaches that deal with different output distributions [7, 8].

A record literature pertains to common random numbers (CRN), in which rival systems are sustained by matched random inputs. Glasserman and Yao [4] recognized CRN as a method of variance reduction when comparing stochastic systems and highlighted that its advantages depend on the dependence structure given by outputs. In a subsequent contribution, Chick and Inoue [5] presented CRN-based selection procedures, while the experiments in disease microsimulation demonstrated that synchronized randomness results in positive correlation and lower variance of differences between systems. Thus, both the variance identity based on pairing and the common idea of statistically valid system selection are established outcomes.

Thus, the exact research gap of the present study pertains to the existent research on blackjack which seeks to address the question of which strategy yields the highest expected return. In contrast, classical research in the area of ranking and selection essentially concentrates on maximizing the selected system by employing appropriate selection procedures. In the present research, the established tools for probability theory and variance reduction considering the so-called finite sample misranking parameter as follows: when comparing two random processes with known or estimated expected rankings, one needs to know what probability corresponds to a situation in which a sample of compounds is taken while reversing their rankings, what is the effect of matched evaluation on that value, and how much evidence is sufficient to lower it to some borderline.

3. Definitions and Core Mathematical Setup

Let A and B denote two stochastic decision rules evaluated under random conditions. Let X_i represent the outcome obtained by applying rule A in observation i , and let Y_i represent the corresponding outcome obtained by applying rule B .

To compare the two rules directly, define the paired difference

$$D_i = X_i - Y_i. \quad (1)$$

The expected value of this difference is

$$\mu_D = \mathbb{E}[D_i] = \mathbb{E}[X_i - Y_i]. \quad (2)$$

Throughout this study, the rules are labelled such that

$$\mu_D > 0, \quad (3)$$

so that rule A is the stronger rule in expectation. This does not mean that A must outperform B in every observation. Individual values of D_i may still be positive or negative because of random variation.

For a finite experiment containing n observations, the mean paired difference is

$$\bar{D}_n = \frac{1}{n} \sum_{i=1}^n D_i. \quad (4)$$

The sign of $\bar{D}_n$ determines the ranking produced by the observed sample. If

$$\bar{D}_n > 0, \quad (5)$$

the observed ranking agrees with the expected ranking. However, despite $\mu_D > 0$, random variation may produce

$$\bar{D}_n < 0. \quad (6)$$

This event is defined in this study as a **strict finite-sample reversal**, or finite-sample misranking. Its probability at sample size n is therefore

$$P_{\text{rev}}(n) = \mathbb{P}(\bar{D}_n < 0), \quad \mu_D > 0. \quad (7)$$

A value of $\bar{D}_n = 0$ is treated as a tie rather than a strict reversal.

The difficulty of distinguishing the two rules depends not only on the size of the expected advantage, but also on the variability of the paired differences. Let

$$\sigma_D^2 = \text{Var}(D_i), \quad (8)$$

and define the standardized effect

$$\delta = \frac{\mu_D}{\sigma_D}. \quad (9)$$

The variable δ exhibits the anticipatory gain in comparison to noise. Higher values of δ mean the better rule is easier to be identified. Lower values of δ mean that more evidence may be necessary for identification.

These definitions accomplish the task of reducing the comparisons between two stochastic theories to studying the behavior of a single random variable D_i . The following sections examine how the fluctuations of D_i and the size of the sample n influence the chances of making a wrong ranking in a finite sample.

The mathematical derivations in this study are formulated for the comparison of two stochastic strategies at a time. This pairwise formulation is the basic unit of PFME and can be applied to any selected pair of strategies. In the blackjack experiments, seven strategies are simulated for broader comparative and robustness analysis, while the finite-sample misranking calculations are evaluated through pairwise comparisons rather than as a simultaneous seven-strategy selection procedure.

4. Paired Variance Reduction

If both variables tend to exhibit similar variations, then the product of their deviations from their respective means will usually be positive, and therefore the covariance will also tend to be positive. If, on the other hand, they vary in opposite directions, the covariance will tend to be negative. This explains the meaning of covariance in experimental testing: it indicates the degree to which the two strategies behave similarly under the same randomness.

In matched-pair testing, both strategies are exposed to the same or corresponding random conditions. If favourable conditions improve both strategies, or unfavourable conditions worsen both of them, then their outcomes tend to vary in the same direction, producing positive covariance.

From the variance identity

$$\text{Var}(D_i) = \text{Var}(X_i) + \text{Var}(Y_i) - 2\text{Cov}(X_i, Y_i),$$

the identity can be derived formally from the definition of variance. Let

$$\mu_X = \mathbb{E}[X_i], \quad \mu_Y = \mathbb{E}[Y_i].$$

Since $D_i = X_i - Y_i$,

$$D_i - \mathbb{E}[D_i] = (X_i - \mu_X) - (Y_i - \mu_Y).$$

Therefore,

$$\begin{aligned} \text{Var}(D_i) &= \mathbb{E} \left[((X_i - \mu_X) - (Y_i - \mu_Y))^2 \right] \\ &= \mathbb{E} [(X_i - \mu_X)^2] + \mathbb{E} [(Y_i - \mu_Y)^2] \\ &\quad - 2\mathbb{E} [(X_i - \mu_X)(Y_i - \mu_Y)]. \end{aligned} \tag{10}$$

Using the definitions of variance and covariance,

$$\boxed{\text{Var}(D_i) = \text{Var}(X_i) + \text{Var}(Y_i) - 2\text{Cov}(X_i, Y_i)} \tag{11}$$

positive covariance reduces the variance of the difference between the two strategies.

For an unpaired evaluation, where the two outcomes are generated independently,

$$\text{Cov}(X_i, Y_i) \approx 0,$$

so

$$\text{Var}(D_{i,\text{unpaired}}) \approx \text{Var}(X_i) + \text{Var}(Y_i). \tag{12}$$

In a matched-pair evaluation with positive covariance,

$$\text{Var}(D_{i,\text{paired}}) < \text{Var}(D_{i,\text{unpaired}}). \tag{13}$$

The mechanism can also be understood by separating each outcome into a common environmental component and strategy-specific variation. Suppose

$$X_i = \mu_X + Z_i + E_{X,i}, \quad Y_i = \mu_Y + Z_i + E_{Y,i}, \tag{14}$$

where Z_i represents variation in the stochastic environment shared by both strategies. Subtracting the two outcomes gives

$$\begin{aligned} D_i &= X_i - Y_i \\ &= (\mu_X - \mu_Y) + (E_{X,i} - E_{Y,i}). \end{aligned} \tag{15}$$

The common component Z_i therefore cancels from the paired difference. This illustrates why matched evaluation can reduce comparison variance: variation produced by the shared environment is partially removed when the two outcomes are compared directly.

At the same time, the expected difference remains

$$\begin{aligned} \mathbb{E}[D_i] &= \mathbb{E}[X_i - Y_i] \\ &= \mathbb{E}[X_i] - \mathbb{E}[Y_i] \\ &= \mu_X - \mu_Y. \end{aligned} \tag{16}$$

Therefore, provided that the matched design preserves the marginal behaviour of the two strategies, pairing changes the covariance structure of the comparison rather than the underlying expected advantage.

Pairing therefore does not imply that strategy A is better than strategy B . Instead, it allows the difference between them to be measured more accurately by reducing the amount of shared environmental variation contained in the comparison. The expected advantage remains unchanged; only the variability around that advantage is reduced.

5. Finite-Sample Misranking Probability

Assume the paired differences D_i are approximately independent, with

$$\mathbb{E}[D_i] = \mu_D > 0 \quad (17)$$

and

$$\text{Var}(D_i) = \sigma_D^2. \quad (18)$$

For a sample of n paired observations, the mean paired difference is

$$\bar{D}_n = \frac{1}{n} \sum_{i=1}^n D_i.$$

By linearity of expectation,

$$\mathbb{E}[\bar{D}_n] = \mu_D, \quad (19)$$

while, under approximate independence,

$$\text{Var}(\bar{D}_n) = \frac{\sigma_D^2}{n}. \quad (20)$$

Thus, increasing the sample size does not change the expected advantage μ_D , but reduces the variance of the observed mean difference at the rate $1/n$.

By the Central Limit Theorem, for sufficiently large n , the sampling distribution of $\bar{D}_n$ can be approximated by

$$\bar{D}_n \approx \mathcal{N}\left(\mu_D, \frac{\sigma_D^2}{n}\right). \quad (21)$$

A finite-sample misranking occurs when the observed mean difference becomes negative even though $\mu_D > 0$. Therefore,

$$\mathbb{P}(\bar{D}_n < 0) = \mathbb{P}\left(\frac{\bar{D}_n - \mu_D}{\sigma_D/\sqrt{n}} < \frac{0 - \mu_D}{\sigma_D/\sqrt{n}}\right) \quad (22)$$

$$= \mathbb{P}\left(Z < -\frac{\sqrt{n}\mu_D}{\sigma_D}\right), \quad (23)$$

where $Z \sim \mathcal{N}(0, 1)$. Hence,

$$\boxed{\mathbb{P}(\bar{D}_n < 0) \approx \Phi\left(-\frac{\sqrt{n}\mu_D}{\sigma_D}\right)} \quad (24)$$

where Φ is the cumulative distribution function of the standard normal distribution.

Using the standardized effect

$$\delta = \frac{\mu_D}{\sigma_D},$$

the expression becomes

$$\boxed{P_{\text{rev}}(n) \approx \Phi(-\sqrt{n}\delta)} \quad (25)$$

This relationship shows that the probability of finite-sample misranking is controlled by three quantities: the sample size n , the true mean advantage μ_D , and the variability of the paired differences σ_D . A larger μ_D moves the expected difference farther away from the reversal boundary at zero, making misranking less likely. A larger σ_D increases the spread of the sampling distribution, making reversal more likely. Increasing n reduces the standard error from σ_D to $\sigma_D/\sqrt{n}$ and therefore makes the observed ranking progressively more reliable.

The effect of sample size can be seen directly from the argument of Φ . Since $\mu_D > 0$ and $\sigma_D > 0$,

$$-\frac{\sqrt{n}\mu_D}{\sigma_D}$$

becomes more negative as n increases. Because $\Phi(x)$ is an increasing function of x ,

$$P_{\text{rev}}(n) \longrightarrow 0 \quad \text{as } n \longrightarrow \infty. \quad (26)$$

Therefore, the probability that the inferior strategy appears superior decreases as the sample size grows. Importantly, however, the rate of this decrease depends on the ratio μ_D/σ_D . A small true advantage hidden within highly variable outcomes may require a large sample before the probability of misranking becomes acceptably low, whereas a larger advantage with lower comparison variance can be identified reliably with substantially fewer observations.

6. Required Sample Size and Advanced Bounds

The previous section gives the normal approximation

$$P_{\text{rev}}(n) \approx \Phi\left(-\frac{\sqrt{n}\mu_D}{\sigma_D}\right).$$

Rather than asking for the reversal probability at a fixed sample size, this expression can be inverted to determine how much evidence is required for a chosen reliability level.

Required sample size. Let α be the maximum acceptable probability of finite-sample misranking. Requiring

$$P_{\text{rev}}(n) \leq \alpha$$

gives, under the normal approximation,

$$\Phi\left(-\frac{\sqrt{n}\mu_D}{\sigma_D}\right) \leq \alpha.$$

Because Φ is increasing,

$$-\frac{\sqrt{n}\mu_D}{\sigma_D} \leq \Phi^{-1}(\alpha) = -z_{1-\alpha},$$

and therefore

$$\frac{\sqrt{n}\mu_D}{\sigma_D} \geq z_{1-\alpha}.$$

Squaring both sides gives

$$n_\alpha = \left\lceil \left(\frac{z_{1-\alpha}\sigma_D}{\mu_D} \right)^2 \right\rceil \quad (27)$$

where the ceiling function is used because the sample size must be an integer. Using

$$\delta = \frac{\mu_D}{\sigma_D},$$

this becomes

$$n_\alpha = \left\lceil \left(\frac{z_{1-\alpha}}{\delta} \right)^2 \right\rceil. \quad (28)$$

Ignoring integer rounding,

$$n_\alpha \approx \frac{z_{1-\alpha}^2}{\delta^2},$$

so

$$n_\alpha \propto \frac{1}{\delta^2}. \quad (29)$$

This inverse-square relationship is important. If the standardized effect δ is halved, the required sample size increases by approximately a factor of four. Conversely, reducing σ_D while leaving μ_D unchanged increases δ and reduces the amount of evidence required.

This connects directly to pairing. If

$$\sigma_{D,\text{paired}} < \sigma_{D,\text{unpaired}},$$

then

$$\delta_{\text{paired}} > \delta_{\text{unpaired}},$$

and consequently

$$n_{\alpha,\text{paired}} < n_{\alpha,\text{unpaired}}.$$

For the same μ_D and α ,

$$\frac{n_{\alpha,\text{paired}}}{n_{\alpha,\text{unpaired}}} \approx \frac{\sigma_{D,\text{paired}}^2}{\sigma_{D,\text{unpaired}}^2}. \quad (30)$$

Thus, variance reduction from pairing can be interpreted directly as a reduction in the required sample size.

Berry–Esseen approximation diagnostic. The normal planning rule relies on the Central Limit Theorem, but the CLT

alone does not state how accurate the approximation is for a finite n . Suppose $D_1, \dots, D_n$ are independent and identically distributed with

$$\mathbb{E}[D_i] = \mu_D, \quad \text{Var}(D_i) = \sigma_D^2,$$

and finite third absolute central moment

$$\rho = \mathbb{E}[|D_i - \mu_D|^3] < \infty.$$

The Berry–Esseen theorem gives

$$\sup_x \left| \mathbb{P} \left(\frac{\sqrt{n}(\bar{D}_n - \mu_D)}{\sigma_D} \leq x \right) - \Phi(x) \right| \leq \frac{C\rho}{\sigma_D^3 \sqrt{n}}, \quad (31)$$

where C is a universal constant.

For the reversal event $\bar{D}_n < 0$, set

$$x_n = -\frac{\sqrt{n}\mu_D}{\sigma_D}.$$

Then

$$\left| P_{\text{rev}}(n) - \Phi \left(-\frac{\sqrt{n}\mu_D}{\sigma_D} \right) \right| \leq \frac{C\rho}{\sigma_D^3 \sqrt{n}}. \quad (32)$$

Hence the possible discrepancy between the true reversal probability and the normal approximation decreases at order

$$O(n^{-1/2}).$$

Berry–Esseen is therefore used in PFME as an approximation-error diagnostic rather than as a replacement for the normal reversal formula. The bound may also be conservative, particularly for discrete or skewed distributions, which motivates comparison with exact finite-sample results where available.

Hoeffding distribution-free bound. When the paired differences are bounded,

$$D_i \in [a, b],$$

and $\mathbb{E}[D_i] = \mu_D > 0$, a reversal implies

$$\bar{D}_n - \mu_D \leq -\mu_D.$$

Hoeffding’s inequality therefore gives

$$P_{\text{rev}}(n) \leq \exp \left(-\frac{2n\mu_D^2}{(b-a)^2} \right). \quad (33)$$

This bound does not rely on approximate normality. It uses only independence, bounded outcomes, the expected advantage, and the range of possible differences.

It can also be inverted. Requiring

$$P_{\text{rev}}(n) \leq \alpha$$

is guaranteed if

$$\exp \left(-\frac{2n\mu_D^2}{(b-a)^2} \right) \leq \alpha.$$

Taking logarithms gives

$$\frac{2n\mu_D^2}{(b-a)^2} \geq \ln \left(\frac{1}{\alpha} \right),$$

and therefore

$$n_{\alpha, \text{Hoeffding}} = \left\lceil \frac{(b-a)^2}{2\mu_D^2} \ln \left(\frac{1}{\alpha} \right) \right\rceil. \quad (34)$$

The inverse-square dependence appears again, since

$$n_{\alpha, \text{Hoeffding}} \propto \frac{1}{\mu_D^2}.$$

The normal approximation, Berry–Esseen theorem, and Hoeffding inequality therefore serve complementary roles within PFME. The normal approximation provides the practical reversal-risk and sample-size estimates; Berry–Esseen diagnoses the possible error of that approximation; and Hoeffding provides a conservative distribution-free bound when the paired differences are bounded.

7. PFME Workflow

7.1 Overview

The Paired Finite-Sample Misranking Estimator (PFME) refers to a series of processes undertaken to measure the reliability with which one stochastic approach can be distinguished from another. It aims to measure the skill in estimating which rule has higher expected return, the likelihood of finite-sample randomness flipping the observed order, as well as the amount of empirical evidence needed to mitigate this risk.

Figure 1 summarizes the workflow.

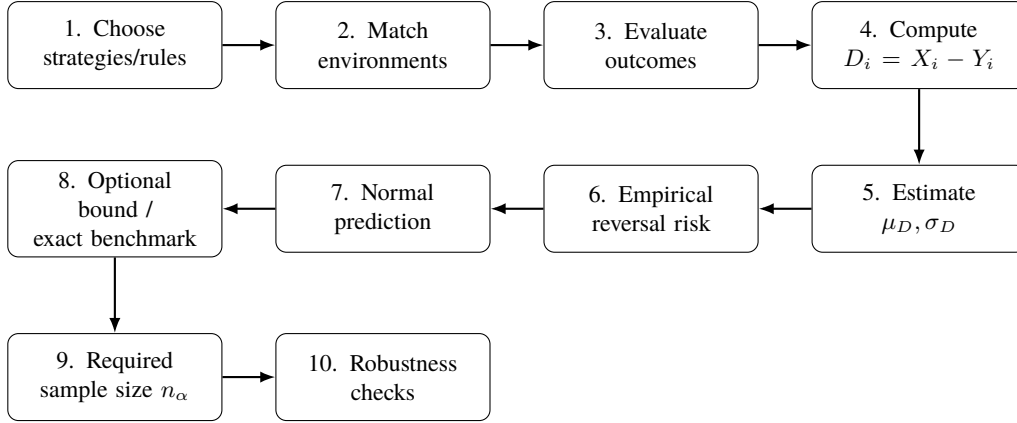


Figure 1: Workflow of the Paired Finite-Sample Misranking Estimator (PFME).

7.2 Ten-Step PFME Procedure

The PFME workflow can be written explicitly as the following ten-step procedure.

1. **Choose the strategies or decision rules.** Select the two rules to be compared. In the pairwise PFME formulation, one rule is labelled A and the other is labelled B .
2. **Match the stochastic environments.** In paired mode, generate the two outcomes under the same or corresponding random environment so that common environmental variation is shared across both rules.
3. **Evaluate the two rules.** For each observation i , compute the outcome of strategy A , denoted X_i , and the outcome of strategy B , denoted Y_i .
4. **Compute the paired difference.** For each matched observation, define

$$D_i = X_i - Y_i.$$

The comparison is then reduced to the behaviour of the single random variable D_i .

5. **Estimate the mean and variability of the comparison.** Compute or estimate the mean paired advantage $\mu_D = \mathbb{E}[D_i]$ and the standard deviation σ_D of the paired differences.
6. **Estimate the empirical reversal probability.** Across repeated finite experiments of size n , estimate how often the observed sample mean $\bar{D}_n$ becomes negative even when $\mu_D > 0$.
7. **Compute the normal-approximation prediction.** Use the PFME approximation

$$P_{\text{rev}}(n) \approx \Phi\left(-\frac{\sqrt{n}\mu_D}{\sigma_D}\right) = \Phi(-\sqrt{n}\delta), \quad \delta = \frac{\mu_D}{\sigma_D}.$$

8. **Optionally compare against a bound or exact benchmark.** Where possible, compare the normal approximation with an exact finite-sample benchmark or with a conservative analytical bound such as Hoeffding's inequality.
9. **Compute the required sample size.** For a target reversal probability α , estimate the required evidence level using

$$n_\alpha = \left\lceil \left(\frac{z_{1-\alpha}\sigma_D}{\mu_D} \right)^2 \right\rceil.$$

10. **Conduct robustness checks.** Repeat the analysis across alternative sample sizes, rule configurations, parameter settings, domains, or benchmarking methods to check whether the conclusions remain stable.

7.3 PFME Outputs

The framework produces several outputs. The most important PFME outputs are:

- the estimated paired advantage μ_D ;

- the estimated comparison variability σ_D ;
- the standardized effect size $\delta = \mu_D / \sigma_D$;
- the empirical reversal probability at each sample size;
- the normal-predicted reversal probability;
- optional exact or bounded reversal-risk benchmarks;
- the required sample size n_α for a chosen reliability target;
- robustness comparisons across domains, rule variants, or modelling assumptions.

Together, these outputs convert a qualitative comparison between strategies into a quantitative statement about decision reliability under finite sampling.

7.4 Paired and Unpaired Modes

PFME can be implemented in either **paired** or **unpaired** mode.

In **paired mode**, the two strategies are evaluated under the same or matched stochastic conditions. This induces covariance between X_i and Y_i , so that

$$\text{Var}(D_i) = \text{Var}(X_i) + \text{Var}(Y_i) - 2\text{Cov}(X_i, Y_i).$$

When the covariance is positive, the variance of the comparison is reduced, which can lower reversal risk and decrease the required sample size.

In **unpaired mode**, the two strategies are evaluated independently. In that case,

$$\text{Cov}(X_i, Y_i) \approx 0,$$

so the variance of the comparison is typically larger:

$$\text{Var}(D_i) \approx \text{Var}(X_i) + \text{Var}(Y_i).$$

The unpaired mode provides the obvious baseline against which pairing efficiency is compared.

The important point to understand is that the process of pairing affects the *variance* of the comparison, but not the expected advantage as such. PFME hence makes a comparison of both modes, thus measuring how much improvement matched evaluation brings to finite sample reliability.

8. Experimental Design and Reproducibility

The PFME framework is evaluated across three stochastic domains: blackjack, inventory control, and a controlled Bernoulli model. These domains were selected because they provide different levels of complexity and analytical tractability. Blackjack serves as the primary high-variance simulation environment; the inventory model provides a structured decision problem with an analytically identifiable superior rule; and the Bernoulli model provides a simple finite-outcome setting in which exact reversal probabilities can be calculated. Together, the three domains allow the framework to be tested using simulation, analytical results, and exact finite-sample benchmarks.

8.1 Simulation Design

For each domain, comparisons are conducted over the sample-size grid

$$n \in \{25, 50, 100, 250, 500, 1000, 2500, 5000, 10000, 25000, 50000\}.$$

Where computationally applicable, each sample-size condition is evaluated over 10,000 repeated experiments. Each repetition produces a finite-sample estimate of the difference between the two strategies and records whether a strict reversal occurs, that is,

$$\bar{D}_n < 0 \quad \text{when } \mu_D > 0.$$

Two evaluation modes are considered. In **paired mode**, both strategies are exposed to the same or corresponding random environment within each comparison. This preserves common stochastic variation and allows covariance between the two outcomes to reduce the variance of their difference. In **unpaired mode**, the strategies are evaluated using independent random environments, providing a baseline against which the efficiency of pairing can be measured.

Random-number generation is controlled so that paired observations share the intended stochastic inputs, while separate Monte Carlo replications remain independently seeded. This distinction is important: common randomness is introduced deliberately *within* a matched comparison, whereas independence is maintained *across* repeated experiments.

For blackjack, direct matched simulation is used through $n = 2500$. Larger sample-size conditions are constructed from matched 2500-round blocks rather than from uninterrupted 50,000-round trajectories. These larger- n results are therefore interpreted as block-based finite-sample estimates rather than as single continuous-shoe simulations.

8.2 Primary Evaluation Metrics

For every applicable domain and sample size, the analysis records the mean outcome of each strategy, the estimated mean difference μ_D , the standard deviation σ_D of the comparison, empirical reversal probability, and the corresponding normal-approximation prediction. Monte Carlo standard errors and confidence intervals are used to distinguish simulation uncertainty from differences between theoretical and empirical results.

The effect of pairing is summarized through the proportional variance reduction

$$VR = 1 - \frac{\text{Var}(D_{\text{paired}})}{\text{Var}(D_{\text{unpaired}})}. \quad (35)$$

The estimated values of μ_D and σ_D are also used to calculate the sample size required for specified target reversal probabilities. Additional sensitivity analysis examines how changes in effect size and comparison variance alter the amount of evidence required.

8.3 Exact Benchmarks and Reproducibility

Where possible, we have used exact computations by hand. The controlled Bernoulli model has a finite paired-difference distribution, and the inventory model allows exact computation of paired reversal probabilities. It enables comparison of the normal PFME estimation with finite-sample ones rather than just Monte Carlo estimates.

The equivalent exact calculation in terms of the full blackjack model is not utilised due to its finite-shoe structure, conditional decisions, splitting, doubling, dealer regulations, and intra-shoe dependence that create a much more complicated process. Thus, blackjack is analysed by means of simulation and approximation mostly.

The computation tool is meant for the reproduction of the stated tests starting with controlled random seeds, preserving the sufficient statistics needed for the metrics reconstruction, running the validation and unit tests, and the reproduction of the main charts and tables. In this respect, it is possible to analyse empirical reversal risk and approximation error separately from the Monte Carlo uncertainty, treating them as one single source of error.

9. Blackjack Model and Simulator Audit

Blackjack serves as the principal high-variance application of PFME. The goal here is not to find an unknown optimal blackjack strategy since a basic strategy for given rules is already known, but instead, to take advantage of blackjack as a controlled stochastic environment for measuring finite-sample ranking errors. The findings reported in this study are generated using the simulator that is explained as follows

9.1 Corrected Blackjack Model

The baseline experiment uses a finite six-deck shoe containing 312 cards. The principal rule configuration is summarized in Table 1.

Table 1: Baseline blackjack simulation rules.

Rule	Baseline setting
Number of decks	6 (312 cards)
Dealer rule	Stand on soft 17 (S17)
Blackjack payout	3:2
Shoe penetration	75%
Double after split	Allowed (DAS)
Surrender	Not allowed
Maximum splits	3
Resplitting aces	Not allowed in baseline

In this process, the cards drawn do not include replacement from the active shoe until a predetermined threshold of penetration has been reached; therefore, at this point, a new shoe of six decks will be generated. The cards used to evaluate the winning or losing hands of the players and dealers will be according to the rules adopted in this game (blackjack treatment, soft-hand treatment, doubling and splitting method, etc).

9.2 Strategies Evaluated

The principal PFME comparison is between **Full Basic Strategy** and an intentionally weaker deterministic **Simplified Strategy**. Full Basic Strategy serves as the established reference strategy under the simulated rule configuration, while the Simplified Strategy provides a lower-performing comparator through which finite-sample reversal can be studied.

To test whether conclusions depend excessively on this single comparison, seven strategies are simulated:

1. Full Basic Strategy;
2. Simplified Strategy;
3. Basic Strategy without splitting;
4. Basic Strategy without doubling;
5. Dealer-mimic strategy;
6. Random legal-action strategy;
7. Conservative stand-on-12 strategy.

The mathematical derivations of PFME remain **pairwise**. The seven-strategy experiment provides a broader comparison and robustness test; reversal probabilities and required sample sizes always refer to a specified pair of strategies rather than to a simultaneous seven-way ranking-and-selection guarantee.

Additional rule variants include changes in deck count, S17 versus H17, 3:2 versus 6:5 blackjack payouts, penetration levels, surrender availability, and resplitting rules.

9.3 Legacy Simulator Audit

During development, an audit identified a structural error in the earlier blackjack implementation. The legacy shoe construction contained only 78 card entries rather than the required $6 \times 52 = 312$ cards for a six-deck game. The penetration logic was nevertheless based on a threshold associated with 78 cards. As a result, the intended finite six-deck shoe was not being represented correctly and the simulator effectively reshuffled far more frequently than intended.

This error materially changes the stochastic structure of the experiment because consecutive blackjack rounds within a finite shoe are dependent through card removal and changing deck composition. The simulator was therefore rebuilt using an explicit 312-card shoe and penetration-based reshuffling.

All corrected results in this study supersede values obtained from the legacy simulator.

Validation and Unit Testing. Before final experiments, the corrected simulator was tested for shoe composition and penetration; hard and soft totals; S17/H17 dealer logic; blackjack payouts; doubling and DAS; splitting, split-ace and resplitting rules; hand settlement; wager accounting; legal strategy actions; and reproducibility under fixed random seeds. These tests distinguish simulator errors from genuine strategy differences and provide the validated computational basis for the blackjack PFME experiments.

10. Blackjack Results

10.1 Finite-Sample Reversal Risk

Let $X_r^{(n)}$ and $Y_r^{(n)}$ denote the mean returns of full basic strategy and simplified strategy in replication r . With $R = 10,000$, empirical reversal risk was estimated by

$$\hat{P}_{\text{rev}}(n) = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{X_r^{(n)} - Y_r^{(n)} \leq 0\}.$$

The corrected six-deck simulator produced long-run mean returns of approximately -0.50% for full basic strategy and -6.08% for the simplified strategy, an expected advantage of approximately 5.58 percentage points. Despite this advantage, finite samples frequently reversed the observed ordering.

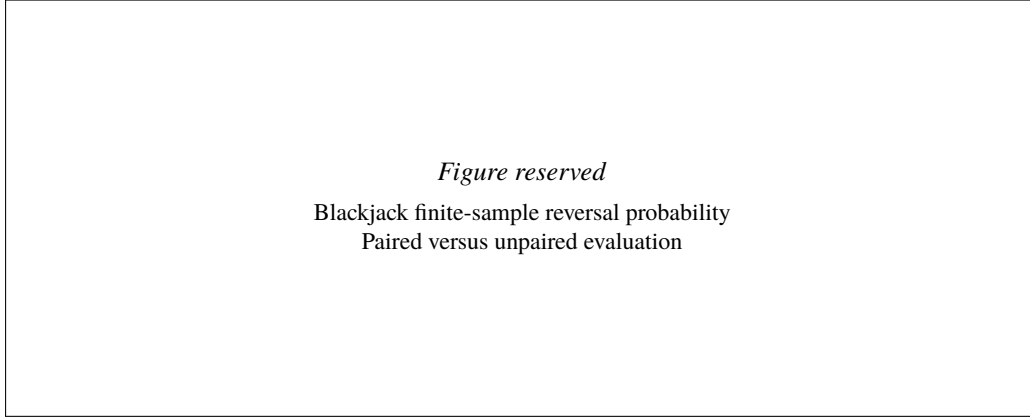


Figure 2: Empirical and normal-approximation blackjack reversal risk under paired and unpaired evaluation.

Table 2: Selected paired blackjack reversal probabilities.

n	Empirical	Normal	95% interval
100	33.29%	33.94%	[32.37, 34.22]%
500	17.37%	17.68%	[16.63, 18.13]%
1,000	9.22%	9.49%	[8.66, 9.80]%
2,500	2.00%	2.01%	[1.73, 2.29]%
5,000	0.19%	0.16%	[0.11, 0.30]%

10.2 Effect of Pairing and Required Evidence

Matched evaluation induced positive covariance and reduced the variance of the strategy difference by a median of approximately 19.6%. At $n = 1,000$, reversal risk was 9.22% under pairing and 11.36% under independent evaluation; at $n = 2,500$, the corresponding values were 2.00% and 3.34%.

Using

$$n_\alpha = \left\lceil \left(\frac{z_{1-\alpha}\sigma_D}{\mu_D} \right)^2 \right\rceil,$$

the estimated sample requirements were

Target risk	Paired n	Unpaired n
10%	974	1,206
5%	1,604	1,987
1%	3,208	3,974
0.1%	5,660	7,012

Thus, pairing saves approximately 766 rounds at the 1% target.

10.3 Robustness and Large- n Results

For a robustness check, apart from the 2 main decision rules compared, 5 other decision rules were added to the PFME. At $n = 2,500$, estimated mean returns were -0.49% for full basic strategy, -1.05% without splitting, -2.01% without doubling, -3.98% for stand-on-12, -5.71% for dealer mimic, -6.10% for simplified strategy, and -42.55% for random legal play. The basic-strategy advantage over simplified strategy remained positive under all tested rule variants.

Conditions above $n = 2,500$ were generated by aggregating independently resampled matched 2,500-round blocks rather than uninterrupted long trajectories. No reversals were observed among 10,000 replications for $n \geq 10,000$; this does not imply zero true risk. With zero observed events, the exact 95% upper confidence bound is approximately 0.0369%.

11. Inventory-Control Validation

11.1 Model and Analytical Optimum

Suppose a firm models how much stock to be ordered before uncertain demand is realized. This was chosen as a case study as optimal order quantity can be derived analytically, while paired evaluation can reuse the same demand realizations. Hence, Inventory control creates a stochastic decision environment in which uncertain demand determines the profit that enables PFME to evaluate its effectiveness in paired comparisons. Suppose Daily demand follows:

$$D \sim \text{Poisson}(20),$$

with profit

$$\pi(q, d) = 12 \min(q, d) - 7q + 2 \max(q - d, 0) - 4 \max(d - q, 0).$$

The underage and overage costs are $C_u = 9$ and $C_o = 5$, giving the critical fractile $9/14 = 0.642857$. Since

$$F_D(20) = 0.559093 < \frac{9}{14} \leq F_D(21) = 0.643698,$$

the optimal integer order quantity is $q^* = 21$. The near-optimal comparator is $q = 20$, with

$$\mathbb{E}[\pi(21, D)] = 76.298815, \quad \mathbb{E}[\pi(20, D)] = 75.126111,$$

and therefore

$$\mu_\Delta = \mathbb{E}[\pi(21, D) - \pi(20, D)] = 1.172704.$$

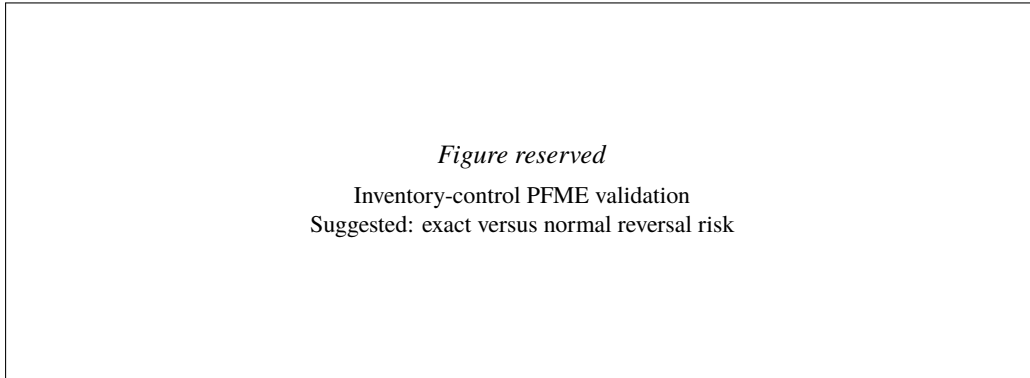


Figure 3: Inventory-control validation of PFME under paired and unpaired evaluation.

11.2 Pairing Through Common Demand

Using the same demand realization for both order quantities gives

$$\Delta(d) = \pi(21, d) - \pi(20, d) = \begin{cases} -5, & d \leq 20, \\ 9, & d \geq 21. \end{cases}$$

The relevant moments are

$$\text{Var}(\pi_{21}) = 647.7246, \quad \text{Var}(\pi_{20}) = 491.3219, \quad \text{Cov}(\pi_{21}, \pi_{20}) = 545.3655.$$

Hence,

$$\text{Var}(\Delta) = 48.3156, \quad \text{Var}(\Delta_{\text{unpaired}}) = 1139.0465,$$

so that

$$VR = 1 - \frac{48.3156}{1139.0465} = 0.9576.$$

Thus, common-demand pairing removes approximately 95.8% of comparison variance.

11.3 Exact Versus Normal Risk

Since

$p_h = P(D \geq 21) = 0.440907$,
and $\Delta_i \in \{-5, 9\}$, if $K \sim \text{Binomial}(n, p_h)$, then

$$\sum_{i=1}^n \Delta_i = 14K - 5n.$$

Therefore,

$$P_{\text{rev}}^{\text{exact}}(n) = P\left(K \leq \left\lfloor \frac{5n}{14} \right\rfloor\right).$$

Table 3: Normal planning and exact paired inventory risk.

Target	Normal n	Normal risk	Exact risk
5%	96	4.9163%	5.2844%
1%	191	0.9860%	1.0530%

Normal planning therefore gives useful sample-size estimates, but not exact finite-sample guarantees. For comparison, the 5% target requires approximately 2,241 observations under unpaired evaluation but only 96 under paired evaluation.

12. Controlled Bernoulli Exact Benchmark

12.1 Model Construction

The Bernoulli case study employs a pair of straightforward success/failure implementations functioning in either simple or difficult environments. We selected this case study because the paired differences are characterized by a well-defined discrete distribution that allows calculating the exact finite sample reversal probabilities.

The Bernoulli model presents a stochastic model of an extremely simple kind, whereby its finite sample reversal probabilities can be calculated with precision.

Let the environment be easy or hard with equal probability. Conditional on the common environment:

	$E = \text{easy}$	$E = \text{hard}$
$P(X = 1 \mid E)$	0.80	0.40
$P(Y = 1 \mid E)$	0.70	0.40

where X represents Rule A and Y Rule B. Hence,

$$\mathbb{E}[X] = 0.60, \quad \mathbb{E}[Y] = 0.55, \quad \mu_D = 0.05.$$

The marginal means are unchanged by pairing; only the dependence structure changes.

12.2 Paired Difference Distribution

For

$$D_i = X_i - Y_i \in \{-1, 0, +1\},$$

the paired distribution is

$$P(D_i = +1) = 0.24, \quad P(D_i = -1) = 0.19, \quad P(D_i = 0) = 0.57.$$

Therefore,

$$\mathbb{E}[D_i] = 0.05, \quad \sigma_D^2 = 0.24 + 0.19 - (0.05)^2 = 0.4275.$$

Independent evaluation gives comparison variance 0.4875, so pairing reduces variance by

$$1 - \frac{0.4275}{0.4875} = 0.1231,$$

or approximately 12.3%.

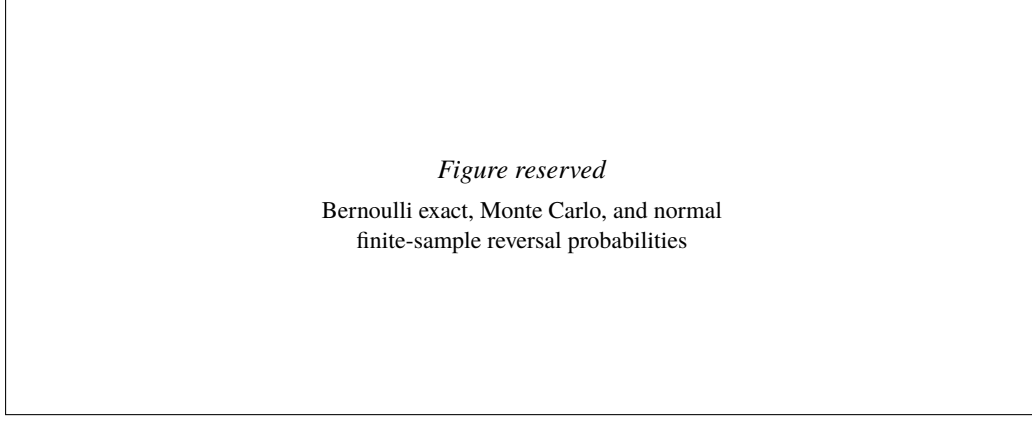


Figure 4: Exact, Monte Carlo, and normal-approximation reversal risk in the controlled Bernoulli benchmark.

12.3 Exact Finite-Sample Benchmark

Let N_+ , N_- , and N_0 denote the numbers of $+1$, -1 , and zero differences. Then

$$(N_+, N_-, N_0) \sim \text{Multinomial}(n; 0.24, 0.19, 0.57),$$

and

$$\sum_{i=1}^n D_i = N_+ - N_-.$$

Thus,

$$P_{\text{rev}}^{\text{exact}}(n) = \sum_{\substack{a, b \geq 0 \\ a+b \leq n \\ a \leq b}} \frac{n!}{a! b! (n-a-b)!} (0.24)^a (0.19)^b (0.57)^{n-a-b}.$$

For independent evaluation,

$$S_X \sim \text{Binomial}(n, 0.60), \quad S_Y \sim \text{Binomial}(n, 0.55),$$

independently, and

$$P_{\text{rev,unpaired}}^{\text{exact}}(n) = P(S_X \leq S_Y).$$

Table 4: Exact, Monte Carlo, and normal Bernoulli reversal risk.

n	Paired exact	Monte Carlo	Normal	Unpaired exact
25	40.859%	41.006%	35.110%	41.421%
50	33.224%	33.265%	29.434%	34.229%
100	24.545%	24.546%	22.222%	25.945%
250	12.286%	12.351%	11.331%	13.854%
500	4.693%	4.738%	4.364%	5.836%
1,000	0.835%	0.824%	0.780%	1.252%

Monte Carlo estimates closely follow the exact probabilities, supporting the implementation. The normal approximation is less accurate at small n , where discreteness and the large probability mass at zero are important, but improves as sample size increases. Pairing consistently reduces reversal risk because the shared environment removes part of the common stochastic variation.

13. Sequential Extension: Brownian Motion and Jump Diffusion

The PFME developed above evaluates misranking at a fixed sample size n . A natural extension is to study how the ranking evolves *during* an experiment. Define the cumulative paired advantage after n observations as

$$S_n = \sum_{i=1}^n D_i. \quad (36)$$

If the paired differences have mean μ_D and variance σ_D^2 , then for sufficiently large n ,

$$S_n \approx n\mu_D + \sigma_D\sqrt{n}Z, \quad Z \sim \mathcal{N}(0, 1). \quad (37)$$

This suggests a continuous-time diffusion approximation

$$\boxed{S(t) = \mu_D t + \sigma_D W(t)}, \quad (38)$$

where $W(t)$ is standard Brownian motion. The drift $\mu_D t$ represents the systematic advantage of the superior strategy, whereas $\sigma_D W(t)$ represents accumulated stochastic noise.

At a fixed time $t = n$,

$$S(n) \sim \mathcal{N}(n\mu_D, n\sigma_D^2),$$

and therefore

$$\mathbb{P}(S(n) < 0) = \Phi\left(-\frac{\sqrt{n}\mu_D}{\sigma_D}\right). \quad (39)$$

Thus, the Brownian model reproduces the normal PFME reversal formula at a fixed observation horizon. Its additional value is that it permits *sequential* questions. For example,

$$\mathbb{P}\left(\inf_{0 \leq t \leq T} S(t) < 0\right) \quad (40)$$

asks for the probability that the inferior strategy appears superior at *any point* before time T , rather than only at the final sample size. Similarly, first-passage times could measure how long it takes for accumulated evidence to cross a chosen reliability boundary.

13.1 Jump-Diffusion Extension

Brownian motion assumes that evidence changes through many small fluctuations. Some stochastic systems, however, may also contain rare but unusually large shocks. To represent this, PFME can be extended using a compound-Poisson jump diffusion:

$$\boxed{S_J(t) = \mu_D t + \sigma_D W(t) + \sum_{k=1}^{N(t)} J_k - \lambda t m_J}, \quad (41)$$

where $N(t)$ is a Poisson process with intensity λ , J_k denotes the size of the k th jump, and

$$m_J = \mathbb{E}[J_k].$$

The compensating term $\lambda t m_J$ centres the jump component, so that the expected cumulative advantage remains

$$\mathbb{E}[S_J(t)] = \mu_D t. \quad (42)$$

If the jumps have second moment $\mathbb{E}[J_k^2]$, then

$$\text{Var}(S_J(t)) = \sigma_D^2 t + \lambda t \mathbb{E}[J_k^2]. \quad (43)$$

As a result, the variable jump makes an additional contribution to opportunity uncertainty on top of the normal sampling noise. Hence, if a strategy is likely to have a positive expected value, it can also be placed in the wrong place in the rankings, owing to either the presence of many minor fluctuations working against it or a rare event producing an unexpected turnaround.

For a Merton-type specification in which

$$J_k \sim \mathcal{N}(m_J, s_J^2),$$

conditioning on $N(t) = m$ gives a normal distribution. Hence the fixed-time reversal probability becomes a Poisson

mixture:

$$P_{\text{rev}}^{\text{jump}}(t) = \sum_{m=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^m}{m!} \Phi \left(-\frac{\mu_D t + (m - \lambda t)m_J}{\sqrt{\sigma_D^2 t + m s_J^2}} \right). \quad (44)$$

This expression reduces to the Brownian PFME approximation when the jump intensity approaches zero.

13.2 Relevance to PFME

The diffusion interpretation expands PFME from a constant-sample question to a sequential one, whereas the jump-diffusion model presents the potential for misranking caused by rare shocks. This extension can be more relevant in fields such as demand, inventory, reliability, or finance, where deranged events may occur.

The empirical findings of the present study do not utilise these models and, therefore, they are suggested as potential extensions of PFME and not as validated elements of the framework. Their significance is to be conceptualised. Currently, the framework seeks to find answers in regard to the final ranking at the sample size n and the possible fact that it may fail, whereas diffusion-PFME can provide knowledge about ranking errors in reverse order, how long the wrong rankings endure and how rare shocks inflict changes on the evidence volume.

14. Discussion

14.1 Interpretation of PFME

The results across blackjack, inventory control, and the Bernoulli benchmark imply the conclusion: the reliability of a stochastic ranking of strategies depends not only on which strategy is better in expectation, but also on the size of that advantage relative to the variability of the comparison. A superior strategy may therefore appear inferior in a finite-sample experiment when the mean advantage is small compared with the noise in the observed difference of the strategies.

PFME expresses this idea through the standardised effect

$$\delta = \frac{\mu_D}{\sigma_D},$$

and the approximate reversal relationship

$$P_{\text{rev}}(n) \approx \Phi(-\sqrt{n} \delta).$$

This makes the roles of effect size, variance, and sample size explicit. Larger μ_D , smaller σ_D , and thus larger sample sizes all make finite-sample reversal less likely.

The paired experiments further show that reliability can be improved without changing the expected advantage itself. When matched stochastic conditions induce positive covariance between the two strategies, the overall variance of their difference, σ_D^2 , decreases. This raises the effective signal-to-noise ratio and can reduce the number of observations required for a target reversal probability. Pairing should therefore be interpreted as a method for improving the precision of comparison, rather than as a method for changing the strategy ranking.

The clearly defined Bernoulli and inventory reference points also shed light on the significance of the normal approximation. While the PFME normal formula can be helpful for analysis and planning purposes, it is not an impeccable finite-sample assurance. Discrepancies between the normal predictions and the finite probabilities are likely in case of small sample sizes, discrete distributions, and/or non-normal distribution of the sample.

14.2 Contribution and Novelty

The mathematical standards used in PFME have already been developed. This work does not re-derive the Central Limit Theorem, covariance formulas, the idea of shared random numbers, Berry-Esseen constraints, or Hoeffding's inequality.

The work adds value by connecting methods for a narrow finite-sample decision problem. PFME considers the probability of *finite-sample misranking* as the key indicator and links this probability to effect size and comparison variance while cross-transforming it into a sample-size rule, assessing the impact of paired versus unpaired experimental design.

The approach is put to the test on three diverse stochastic cases and is not limited to blackjack as its case. Blackjack is a high-variance application that is complex, inventory control is analytically clear, while Bernoulli model is a perfect finite-sample reference. This mix enables to measure empirical simulation error, approximation error, and influence of pairing.

Thus, the novelty claimed in this study is not a new solution to blackjack or a new probability theorem, but a unified finite-sample framework for asking a practical question: *how much evidence is required before the observed ranking of two stochastic strategies can be trusted?*

14.3 Limitations

1. The normal approximation may be inaccurate at small sample sizes, particularly when the paired-difference distribution is skewed or discrete.
2. Blackjack observations within a finite shoe are not fully independent because cards are removed without replacement

and deck composition changes between rounds.

3. Large- n blackjack conditions above 2500 rounds use matched block construction rather than uninterrupted trajectories, and should therefore be interpreted accordingly.
4. Pairing is beneficial only when the matching structure produces useful positive covariance. Poorly chosen or negatively correlated matching can provide little benefit or may increase comparison variance.
5. PFME evaluates the reliability of a specified comparison. It does not by itself prove that a strategy is globally optimal among every possible decision rule.

15. Conclusion

16. Future Work

17. Code and Data Availability

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